

Open Source Log Management Home-Lab using Graylog

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OBJECTIVE

The objective of building an Open Source Log Management Home-Lab using Graylog is to create a fully functional centralized logging environment where logs from multiple systems are collected, parsed, enriched, stored, monitored, and analyzed through Graylog along with its supporting components like MongoDB and OpenSearch, enabling hands-on experience in configuring log inputs, pipelines, alerts, dashboards, retention policies, and real-world scenarios for threat hunting, troubleshooting, auditing, and understanding enterprise-level log management workflows in a safe self-hosted environment.

TOOLS USED

- .Ubuntu Desktop
- .Ubuntu Server
- .VirtualBox
- .Docker
- .Graylog
- .mongoDB
- .Elasticsearch
- .Syslog

Setting up Graylog using Docker on Ubuntu Server

```
gokul@gokul:/$ sudo apt install -y docker.io docker-compose
[sudo] password for gokul:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd pigz python3-docker python3-dockerpty
  python3-dockerpty python3-dockerpty python3-dotenv python3-texttable python3-websocket runc
  ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx
  docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd docker-compose docker.io pigz python3-docker
  python3-dockerpty python3-dockerpty python3-dotenv python3-texttable
  python3-websocket runc ubuntu-fan
0 upgraded, 13 newly installed, 0 to remove and 53 not upgraded.
Need to get 76.2 MB of archives.
After this operation, 289 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 pigz amd64 2.6-1
    [63.6 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 bridge-utils amd64 1
    .7-1ubuntu3 [34.4 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 runc amd64 1
    .3.3-0ubuntu1~22.04.3 [8,857 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 containerd a
```

I set up Graylog using Docker on Ubuntu because Docker provides an easier, faster, and cleaner deployment method where Graylog, MongoDB, and OpenSearch can run as isolated containers without manual installation dependencies, making the environment more portable, scalable, and simple to manage or recreate for learning, testing, and home-lab experiments.

Command: `sudo apt install -y docker.io docker-compose`

```
gokul@gokul:/$ sudo systemctl enable --now docker
gokul@gokul:/$
```

I ran `sudo systemctl enable --now docker` to start the Docker service immediately and also configure it to start automatically on every system boot, ensuring that my Docker-based Graylog environment runs reliably without needing to manually start Docker each time the server restarts.

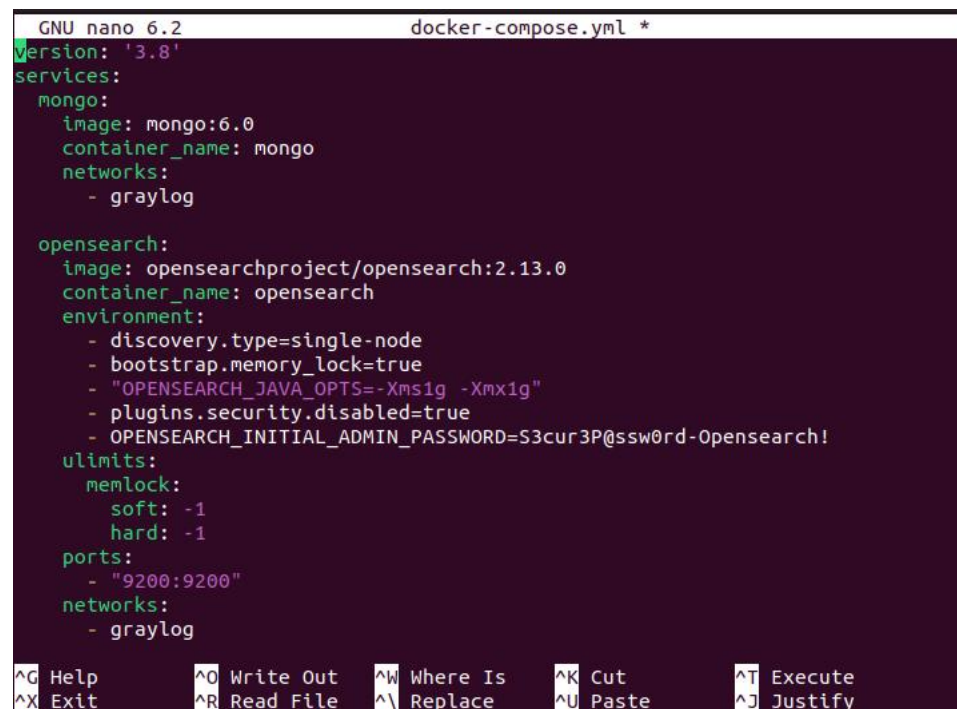
Create directories for Graylog deployment

```
gokul@gokul:/$ sudo mkdir -p /opt/graylog/{mongo,elasticsearch,graylog}
gokul@gokul:/$
```

I created the directories `/opt/graylog/{mongo,elasticsearch,graylog}` and navigated into `/opt/graylog` to organize persistent storage locations for MongoDB, Elasticsearch/OpenSearch, and Graylog data, ensuring that when the Docker containers run they can save all logs, configurations, and database files to these dedicated folders so that my Graylog environment remains stable and retains data even after container restarts.

Command: `sudo mkdir -p /opt/graylog/{mongo,elasticsearch,graylog}`
`cd /opt/graylog`

Create a docker-compose.yml file



```
GNU nano 6.2                docker-compose.yml *
Version: '3.8'
services:
  mongo:
    image: mongo:6.0
    container_name: mongo
    networks:
      - graylog

  opensearch:
    image: opensearchproject/opensearch:2.13.0
    container_name: opensearch
    environment:
      - discovery.type=single-node
      - bootstrap.memory_lock=true
      - "OPENSEARCH_JAVA_OPTS=-Xms1g -Xmx1g"
      - plugins.security.disabled=true
      - OPENSEARCH_INITIAL_ADMIN_PASSWORD=S3cur3P@ssw0rd-Opensearch!
    ulimits:
      memlock:
        soft: -1
        hard: -1
    ports:
      - "9200:9200"
    networks:
      - graylog
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

I created this Docker Compose configuration to automatically deploy a fully functional Graylog stack—consisting of MongoDB for storing Graylog configurations, OpenSearch for indexing and searching log data, and Graylog itself for log collection and analysis—while defining their environment settings, ports, dependencies, networks, and persistent structure so that the entire log management system runs smoothly, consistently, and easily on Ubuntu with a single `docker-compose up -d` command.

```

gokul@gokul:/opt/graylog$ sudo docker-compose up -d
Pulling mongo (mongo:6.0)...
6.0: Pulling from library/mongo
af6eca94c810: Pull complete
1ad79419bb76: Pull complete
afdef96060ff: Pull complete
dc9abaf2c8ff: Pull complete
a2be9b1fc57a: Pull complete
9d81a4e85c6a: Pull complete
9d6e74542339: Pull complete
cde22982277a: Pull complete
Digest: sha256:0a83b2f824b29c94c5284a49b018db2e3166d9879d1edda5cce545d899ef2bbb
Status: Downloaded newer image for mongo:6.0
Pulling opensearch (opensearchproject/opensearch:2.13.0)...
2.13.0: Pulling from opensearchproject/opensearch
81e74100301d: Pull complete
a2155cf45863: Pull complete
a52cf78d8de1: Pull complete
4f4fb700ef54: Pull complete
042bd1985482: Pull complete
9a11437f36e1: Pull complete
Digest: sha256:2e954ff0e8c9d0f4868b4818150b3aecc92fbb0cc4a24d00dace38ada227291d
Status: Downloaded newer image for opensearchproject/opensearch:2.13.0
Pulling graylog (graylog/graylog:5.2)...
5.2: Pulling from graylog/graylog
7478e0ac0f23: Pull complete
90a925ab929a: Pull complete
7d9a34308537: Pull complete
80338217a4ab: Pull complete
1a5fd5c7e184: Pull complete
ac051f74d704: Pull complete
4f4fb700ef54: Pull complete
c0632e8a67d9: Pull complete
24a7332dca55: Pull complete
47643bfb4ee4: Pull complete
Digest: sha256:3f4aa0885cd3e30442e5f7e1b2361f57773831bb84515386eb5dfa6f426fc1d7
Status: Downloaded newer image for graylog/graylog:5.2
Creating opensearch ... done
Creating mongo       ... done
Creating graylog     ... done
gokul@gokul:/opt/graylog$

```

I ran `sudo docker-compose up -d` to start all the Graylog, MongoDB, and OpenSearch containers in the background using the Docker Compose file, allowing the entire log management stack to launch automatically and continue running even after I close the terminal.

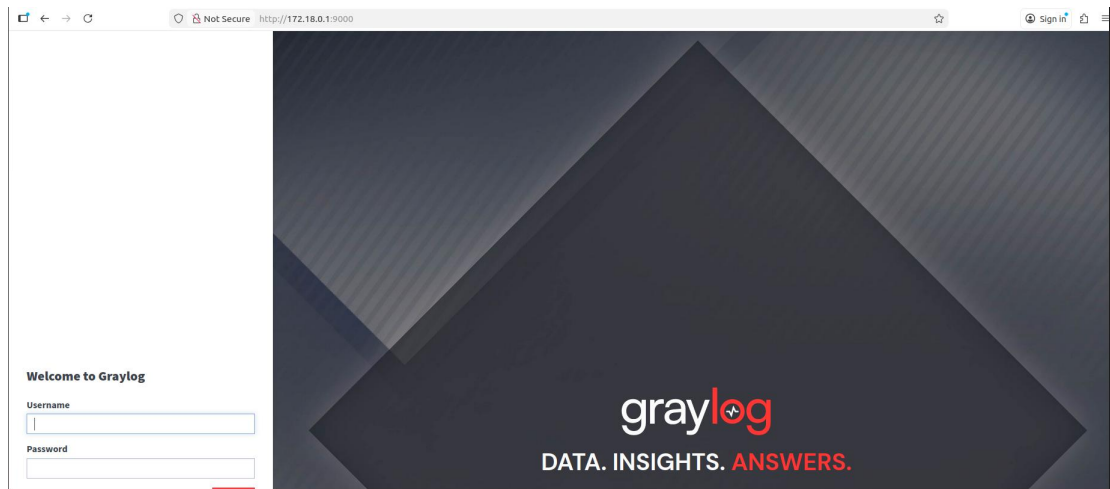
Verify containers are running

```

gokul@gokul:/opt/graylog$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        NAMES                PORTS
c88fba7b0b82   graylog/graylog:5.2                "/usr/bin/tint -- wa..." 58 seconds ago Up 57 seconds (health: starting) 0.0.0.0:1514->1514/udp, [::]:1514->1514/udp, 0.0.0.0:1514->1514/tcp, [::]:1514->1514/tcp
c3b78f06977f   mongo:6.0                          "docker-entrypoint.s..." About a minute ago Up 59 seconds   mongo                27017/tcp
4702b7794728   opensearchproject/opensearch:2.13.0 "/.opensearch-docker..." About a minute ago Up 59 seconds   opensearch           9300/tcp, 9600/tcp, 0.0.0.0:9200->9200/tcp, [::]:9200->9200/tcp, 9650/tcp
gokul@gokul:/opt/graylog$

```

I used `sudo docker ps` to verify that all the Graylog-related containers—Graylog, MongoDB, and OpenSearch—are running correctly after starting them with Docker Compose, ensuring the services are active and healthy.



I accessed <http://myserver ip:9000> to open the Graylog web interface in my browser, allowing me to log in, configure inputs, monitor logs, and manage the entire Graylog system through its graphical dashboard.

Setting up Ubuntu Syslog Client

```
root@ubuntu:~# sudo apt install -y rsyslog
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
rsyslog is already the newest version (8.2112.0-2ubuntu2.2).
rsyslog set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@ubuntu:~#
```

I installed rsyslog (Syslog agent) using `sudo apt install -y rsyslog` to enable my Client Ubuntu system to generate, manage, and forward system logs so that I can send those logs to my Graylog server for centralized log collection and monitoring.

Edit rsyslog configuration to forward logs to the Graylog server

```
root@ubuntu:/# sudo nano /etc/rsyslog.d/90-graylog.conf
root@ubuntu:/#
```

```
GNU nano 6.2 /etc/rsyslog.d/90-graylog.conf
.* @172.18.0.1:1514;RSYSLOG_SyslogProtocol23Format

^G Help      ^O Write Out  ^H Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^A Set Mark
^X Exit      ^R Read File  ^I Replace    ^U Paste      ^J Justify    ^_ Go To Line  ^- Redo       ^- Copy
```

I edited the rsyslog configuration to forward logs to the Graylog server so that all system logs generated on my Ubuntu machine are automatically sent over the network to Graylog, enabling centralized log collection, easier monitoring, searching, and analysis from a

single Graylog web dashboard instead of checking logs locally on each device so I added my ubuntu server ip into configuration file.

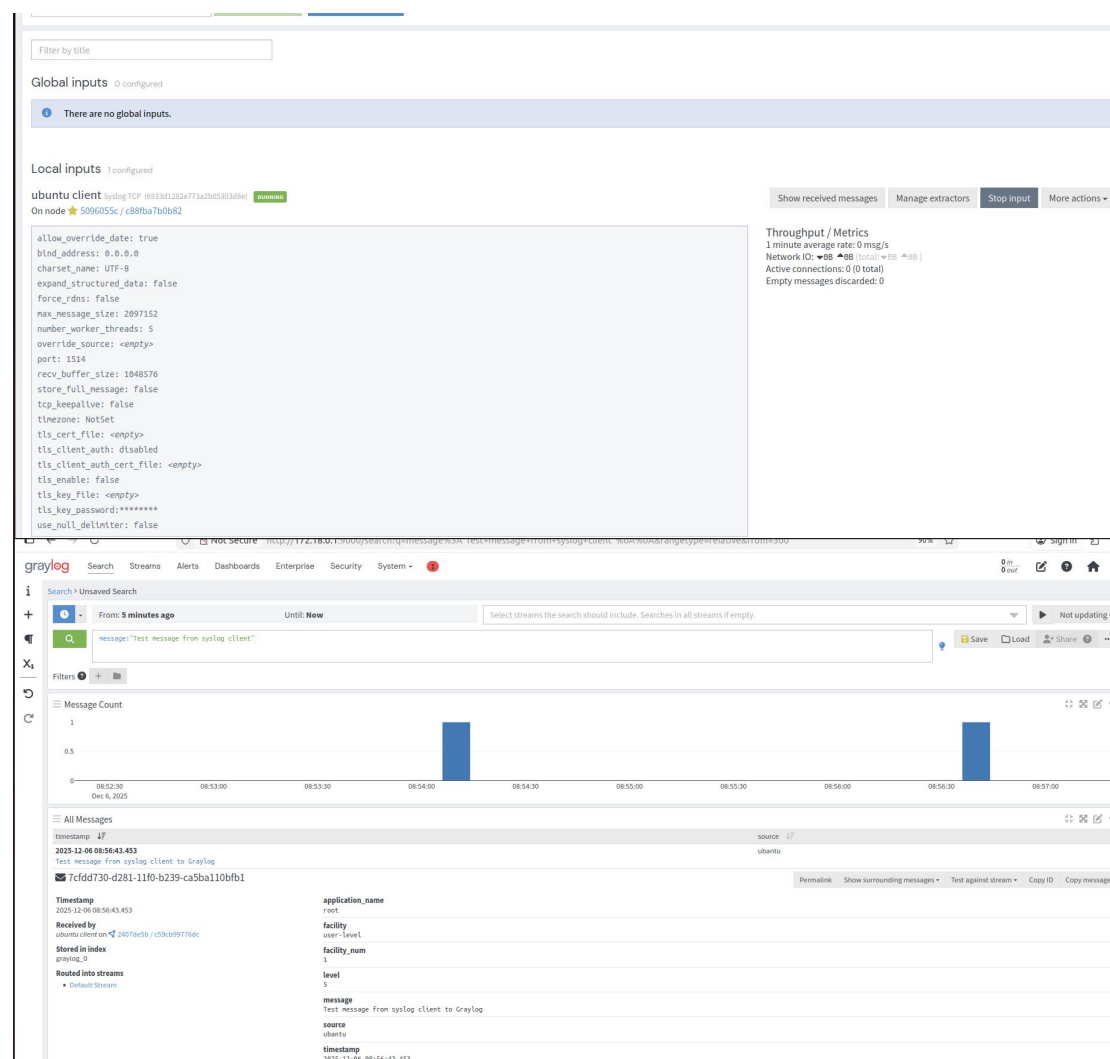
```
root@ubuntu:/# systemctl restart rsyslog
root@ubuntu:/# systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2025-12-06 14:11:54 IST; 19s ago
     TriggeredBy: ● syslog.socket
    Docs: man:rsyslogd(8)
          man:rsyslog.conf(5)
          https://www.rsyslog.com/doc/
   Main PID: 5671 (rsyslogd)
     Tasks: 4 (limit: 14899)
    Memory: 1.1M
       CPU: 121ms
    CGroup: /system.slice/rsyslog.service
            └─5671 /usr/sbin/rsyslogd -n -lNONE

Dec 06 14:11:54 ubuntu systemd[1]: Starting System Logging Service...
Dec 06 14:11:54 ubuntu rsyslogd[5671]: imuxsock: Acquired UNIX socket '/run/systemd/journal/syslog' (fd 3) from systemd. [v8.2112.0]
Dec 06 14:11:54 ubuntu systemd[1]: Started System Logging Service.
Dec 06 14:11:54 ubuntu rsyslogd[5671]: rsyslogd's groupid changed to 111
Dec 06 14:11:54 ubuntu rsyslogd[5671]: rsyslogd's userid changed to 104
Dec 06 14:11:54 ubuntu rsyslogd[5671]: [origin software="rsyslogd" swVersion="8.2112.0" x-pid="5671" x-info="https://www.rsyslog.com"] start
root@ubuntu:/#
```


I ran `sudo systemctl status rsyslog` to check whether the rsyslog service is running properly after configuration changes, ensuring that the log-forwarding agent is active, healthy, and ready to send logs to the Graylog server.

Test log forwarding by creating a test message

```
root@ubuntu:/# logger "Test message from syslog client to Graylog"
root@ubuntu:/# logger "Test message from syslog client to Graylog"
root@ubuntu:/#
```



After sending a test message using the logger command on my Ubuntu client, I searched for that message in Graylog and successfully saw it appear in the Graylog Server, confirming that my syslog client is correctly forwarding logs to the Graylog server.

Conclusion

I have successfully built a centralized open-source log management home lab using Graylog. Logs from your Ubuntu syslog client are now collected, parsed, and visualized in real time. This environment helps simulate how SOC teams aggregate and monitor logs across infrastructure for analysis and alerting.